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Student Success Coaching Is Best Delivered Early and Often

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This research investigates the effects of a success coaching program by examining the timing of the first appointment and the frequency of additional coaching. Using a fuzzy regression discontinuity approach, we provide Local Average Treatment Effects of the program when students make appointments early in the semester. Students with appointments scheduled before discontinuity had an average of two additional meetings compared to students who waited to make their first appointment. We find each coaching session yields an effect of about 0.2–0.3 grade points and approximately 0.8 additional credits earned. This article contributes to understanding of the timing and frequency effects for success coaching and recommends programs aim to have first interactions with students happen before the sixth week of classes and offer four to six appointments per semester for maximum impact.

Institutions of higher education employ various interventions to assist students in their transitions from high school to college, demonstrating the institutional commitment to student well-being (Braxton & Hirschy, 2005; Braxton et al., 2011; Seidman, 2005). Many colleges invest significant resources in high touch interventions for students making the transition from high school to college and use of success coaching is emerging as a potential strategy for promoting student academic success (Ralston & Hoffshire, 2017; Robinson & Gahagan, 2010). However, there is limited rigorous evidence of success coaching interventions, especially when delivered by full-time, professional staff persons employed by a university (Bettinger & Baker, 2014; Oreopoulos & Petronijevic, 2018).

This article contributes to the extant success coaching literature by expanding understandings of the *dosage effects* of success coaching (defined here as the timing and frequency of coaching) and systematically investigating the importance of early interventions on two important student outcomes: first semester grade point average and credits completed. We employ a fuzzy regression discontinuity design, estimating the discontinuity as an instrumental variable to estimate the Local Average Treatment Effect (LATE) of the early interventions. We present

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consistent evidence that there is a benefit to early interventions and that each additional coaching session has substantial benefits for the group that attends early in the semester.

Success Coaching in Higher Education

Student coaching initiatives that aim to facilitate college success and retention have expanded significantly in the last 10 years (Robinson, 2015). Initially rooted in positive psychology and humanistic theories, coaching is a “multidisciplinary, multi-theory synthesis and application of applied behavioral change” that builds upon theories and research from related fields (Williams, 2006, p. 49). Although consistent programmatic elements may not always align, the common thread across many contexts lies in promoting coaching as a holistic approach through which individuals can control and direct their own lives toward fulfillment and maximize their full potential (Jarosz, 2016). Coaching entails a “collaborative, solution-focused, results-oriented and systematic process in which the coach facilitates the enhancement of the coachee” in some aspect of their life (Lai, 2015, p. 13). Coachees are considered creative and resourceful with the ability to change and grow, and coaches provide a judgment-free environment that prioritizes active listening, empowerment, and accountability (Jarosz, 2016). Although originating in business and professional settings, anecdotal evidence related to the effectiveness of coaching in the workplace has demonstrated that coaching programs contribute to improved goal accomplishment and productivity, enhanced professional relationships, and increased resilience and well-being (Jones et al., 2016). Other more generalized outcomes from coaching include the enhancement of one’s sense of self, facilitation of positive behavioral changes, and a better quality of life (Jarosz, 2016).

Before coaching became as distinguished as it is today, institutions implemented early iterations in the form of proactive advising (Butler et al., 2016; Schwebel et al., 2012). Since then, the process has evolved considerably to become more expansive, holistic, and accessible. Within recent years, the practice of coaching has gone mainstream in higher education settings, in large part due to the Bettinger and Baker (2014) study that provided strong evidence for the effectiveness of this intervention on improving student retention and persistence outcomes. Moreover, academically at-risk undergraduates who participated in coaching have seen significant GPA increases and at least a 76% credit completion rate during the intervention semester, with exponential benefits upon attending additional sessions (Capstick et al., 2019). Graduate student outcomes have demonstrated increased likelihood of persistence after just one visit with an academic coach (Lehan et al., 2018). The effectiveness of coaching for underrepresented students of color has pointed to persistence outcomes that exceed those of their majority counterparts, demonstrating the relevance of this intervention for vulnerable populations (Simmons & Smith, 2020).

Distinguishing Success Coaching from Other Interventions. Although many institutions offer similar academic support services designated as tutoring, academic coaching, and academic advising, “success” coaching in this context is specifically distinguished by its holistic approach to student success. In many cases, academic advising still takes on the prescriptive approach of ensuring course registration and curriculum progression; tutoring often functions to demonstrate course-specific cognitive skills, and academic coaching tends to target students at serious academic risk (Capstick et al., 2019; Harris, 2018; Stigmar, 2016; Vanacore & Dahan, 2021). Success coaching, however, can be thought to be more broadly focused, as coaching practices prioritize not just academic success, but also personal well-being. Simmons and Smith (2020) described a model designated as success coaching, which worked to address both the academic

and nonacademic causes of student attrition. Further, success coaching works to promote a wider scope of noncognitive factors, such as sense of belonging, self-efficacy, and perseverance, which play an influential role in student success outcomes, including academic improvement, credit accumulation, and graduation (Farrington et al., 2012).

Dosage Effects: Timing and Frequency of Coaching

Given that success coaching is a relatively new practice, few studies to date have investigated how timing or frequency of success coaching affects students, which can be thought of as kinds of dosage effects. Though “dosage effects” may be a term more frequently used in medicine and epidemiology (Kestenbaum, 2019) than educational research and student affairs, intervention research should be concerned with dose-response relationships as a way of understanding when and how frequently interventions should be administered to achieve the desired effects (Chen et al., 2022). Timing and frequency are among the most commonly researched dosage effects for educational research (de Ree et al., 2023; Reynolds et al., 2011; Turk, 2019). Identifying the dose-response of interventions is conceptually relevant for practitioners, with researchers on coaching initiatives recommending more attention to these kinds of dosage effects of coaching (Alzen et al., 2021; Hall et al., 2021). The extant evidence of dosage effects for success coaching suggests that the frequency and timing of coaching may explain student outcomes ranging from retention, to academic performance, and completion of degree.

In a randomized control trial, Bettinger and Baker (2014) found that a private coaching service yielded an increased retention rate over students not randomly assigned to a coach. This finding was largest when a student responded to outreach from their coach in the first week of classes, supporting the theory that timing matters in regard to success coaching. Additionally, they found the effects of coaching increased as the number of contacts with a coach increased using instrumental variables methods to estimate these dosage effects.

In a large-scale, post-hoc survey of community college students, Pechac and Slantcheva-Durst (2021) found that frequency of coaching was a significant predictor of student credit completion, though most students only reported between one and three sessions. In their report on the Success Coach Boston program (Linkow et al., 2017), the evaluators’ conclusion was that frequent coaching was associated with significantly higher GPA outcomes than students from an adjusted comparison group. The recommended frequency from the Boston program was one interaction per month. Although they use the term “academic coaching” to describe a holistic success coaching model, Alzen et al. (2021) found small but consistent additive effects of additional coaching up to about six coaching sessions.

Another rigorous study that investigated the potential of coaching as a strategy for student success include Oreopoulos and Petronijevic (2018), who found a substantial effect of in-person coaching on earned grades, equal to about half of a letter grade (between 4 and 8 percentage points on a 100-point scale). Their findings were contrasted with a several lower-touch, lower-cost alternatives (such as technological nudges), and demonstrated coaching was more effective despite its increased cost. However, Oreopoulos and Petronijevic (2018) used upper-division students as coaches instead of professionals.

Current Study

This study investigates a coaching program at a medium-sized, high-research activity, urban Minority-Serving university. The student population includes 57% who are first-generation, 52% who are Pell-eligible, and 39% who represent minoritized students. The institution pairs a

professional success coach with first-time in college students during their first semester in college to aid in the transition into college, provide support for students in navigating the institution and develop a sense of belonging. The aim of the study is to better understand how timing and frequency of success coaching matter for program effectiveness. Our research questions are shaped by Bandura's (1997) social cognitive theory (SCT), which adopts an agentic perspective to human development, adaptation, and change. The theory distinguishes among three modes of agency: personal agency exercised individually; proxy agency in which people secure desired outcomes by influencing others to act on their behalf; and collective agency in which people act in concert to shape their future. When coaching occurs consistently, SCT can be demonstrated via increased perceptions of self-efficacy, goal setting, and enhanced outcome expectations (people's belief about the possible consequences of their actions), which lead to desired outcomes. Using an approach similar to Vanacore and Dahan (2021), we operationalize this concept through the timing and frequency of success coaching variable because these characteristics may meaningfully represent the agentic perspectives embedded in SCT (Bandura, 2001) and reflect the enactment of the coaching protocol described here, focused on increasing the agency of students to achieve their goals. Thus, we pose the following research questions: Does timing of the first interaction with a success coach matter to the number of subsequent meetings? To what extent do students who access success coaching early in the semester benefit from their ongoing participation in coaching?

Methods

This section includes study procedure, a statement of researcher positionality, and a detailed description of the coaching program. Summary information regarding the analytic sample and details regarding the empirical strategy using Fuzzy Regression Discontinuity Design to estimate the dosage effects of coaching frequency and timing is provided.

Researcher Positionality

The researchers in this study are committed to equity and student success and offer readers acknowledgment of their positionality in regards to this research (Jamieson et al., 2023; Roberts et al., 2020). The primary author is a White male and served as the director of student success analytics during research. One of the coauthors is a Black female who served as the assistant director of the coaching program, managing the day-to-day operations, at the time the research was conducted. The final coauthor is a Black female who held the role of assistant chancellor for student academic success during the time of this study. Each of these roles influenced how we approached the research. In the area of data collection, we considered multiple identities of students when requesting the deidentified student information used in this study but acknowledge the myriad limitations of how our own institution and the field of higher education records these data. In analysis, we acknowledge our lack of objectivity as we examine our own programming. In interpretation of these data, we recognize that by examining dosage effects, we embed assumptions about how humans form relationships and reduce that complex process to relative dates of first contact and counts of interactions, using this proxy measure in place of other potential operationalizations of the theories we are testing.

Procedure

Here we describe the coaching program procedures before describing the sampling procedures for this study. The purpose is to create the theory-practice linkage between the program

and its outcomes that are examined in the study. Each Fall, all incoming students who were not otherwise enrolled in a program aimed at a special population (e.g., the Educational Opportunity Fund program, TRiO SSS program, or Honors program) were eligible to participate in success coaching. Students were recruited for the coaching program during the summer before their first semester of college via mailed informational materials, text message outreach, e-mails, and during in-person orientation sessions. Appointment campaigns and requests for initial meetings were sent to all eligible students via e-mail on a weekly basis between August and October. Initial appointments began two weeks prior to the start of the fall term and continued throughout the fall semester.

Coaching Model. The coaching model was comprised of three components: 1) individual and group coaching sessions, 2) educational and skill building workshops, and 3) referral and case management services. The primary objectives of the implemented program were to ease the college transition by promoting academic and social skills, proactively addressing barriers to success, and encouraging the establishment of meaningful student relationships with peers, faculty, and staff across campus. Participation in the coaching program was not mandatory; however, it was strongly encouraged during recruitment outreach over the summer and throughout the fall semester. The coaching program was advertised as a one-year commitment on behalf of students, whereby upon completion of the program, students would feel empowered to utilize their expanded networks of support as they progressed toward graduation.

Success Coaches.

Success Coaches included four full-time staff members who each held master's degrees in counseling or higher education. Professional work experience with students in postsecondary education ranged from 4–11 years. All coaches implemented Appreciative Advising techniques in their practice (Bloom et al., 2013; Hutson, 2010). Appreciative Advising is rooted in the tenets of positive psychology, Appreciative Inquiry, and motivational theory. It is a strengths-based, student-centered, and personalized approach to student development that helps students optimize their educational experiences. Associated coaching strategies were focused on building rapport, open-ended questioning, collaboration, and setting high expectations (Bloom et al., 2008). Coaching caseloads were alphabetically assigned based on students' last names, with each coach serving approximately 149 students each.

Coaching Sessions.

Initial coaching sessions began two weeks before the start of the fall semester, with the expectation that meetings would continue on a biweekly basis throughout the remainder of the term. Coaches inquired with students about any challenges or questions pertaining to the success of their first semester and asked students for their feedback regarding their experience thus far. Coaches utilized these sessions to review students' success portfolios, which included critical information regarding the academic calendar, textbooks, term bills, learning management platforms, campus support services, and other ready-to-use resources to aid in their transition to college life. All information was shared via hardcopy or electronically uploaded within the data management system, which students could access at their convenience. In subsequent coach meetings, success coaches engaged in discussions with students regarding their goal achievement, personal strengths and interests, and academic progress. Conversations and activities to promote success strategies related to academic skill building, goal planning, and help-seeking behavior were incorporated in each session. Additionally, sessions were supplemented by monthly e-mail newsletters, weekly text message outreach, and periodic phone call campaigns that nudged

students to act on items such as course registration, progress reports, or appointment requests for support services.

Programming.

Monthly educational and social programming was facilitated by success coaches to further promote students' development of academic and personal skills, and to provide opportunities for students to engage with faculty and peers outside of the classroom environment. Ranging from 30 minutes to an hour, events included student and faculty panels, networking mixers, time management and study skills workshops, and recognition ceremonies to celebrate students' successes. Students who attended reported feedback to verify learning outcomes and evaluate the relevance of each topic to better inform future programming.

Referral and Case Management Services.

In providing holistic services, success coaches functioned as liaisons between students and other campus supports, including the wellness center, learning center, and dean of students office. In response to individualized concerns, coaches provided timely referrals to appropriate support staff to better assist with students' needs. Coaches regularly monitored cases throughout the resolution process and intervened with additional support when necessary. By proactively and promptly addressing barriers to success, coaches helped students prevent, prepare for, and effectively respond to common obstacles experienced in higher education.

Participants

The sample in this study includes 175 students receiving services as part of the success coaching program who agreed to participate in the study. At each student's first coaching session, their coach provided informed consent regarding the research activities and students had the option to volunteer to participate. This study qualified as volunteer participation in research for introductory Psychology students but was open to all students served by the Success Coaching Office to help continuously assess and improve coaching practices. The study participants represent about half of the students served by the coaching program during the semester we conducted this research because some students declined consent to participation.

The average age of participants was 18.1 years old, with 32.4% recorded as male and 67.6% recorded as female. The sample reflected the diverse population of the campus—White (37.6%), African American (20.8%), Hispanic (18.5%), Asian (12.7%), and other (10.4%). All participants were first-time, first-year undergraduates, and 53.2% were first-generation students. [Table 1](#) includes descriptive statistics of this student population.

Research Design

The empirical strategy of this study uses a fuzzy regression discontinuity design based on an exogenously determined cutoff point in the semester to define an early intervention and identify the effects of additional student success coaching sessions on student GPA and credits completed. In the regression discontinuity literature, a "running variable" is selected that is related to the variable of interest but features a cutoff discontinuity outside of the researchers' and the subject's control. In true regression discontinuities designs, the discontinuity is deterministic, such as a test score threshold for admission to a scholarship program (Thistlethwaite & Campbell, 1960).

Table 1

Sample Summary Statistics

	Mean	SD	Min	Max
<i>Outcome Variables</i>				
Term GPA	3.127	.791	0	4
Term credits completed	15.309	2.532	0	22
<i>Covariates</i>				
Pell Indicator	.486	.501	0	1
African American	.208	.407	0	1
Asian	.127	.334	0	1
Hispanic	.185	.389	0	1
Other	.104	.306	0	1
Gender = Female	.676	.469	0	1
First Gen	.532	.501	0	1
Mean-centered SAT Score (divided by 100)	.005	1.130	-3.167	2.634
Standardized values of high school GPA	-.010	.999	-2.193	2.412
Indicator for missing SAT = 1	.040	.198	0	1
Indicator for missing HS GPA = 1	.023	.151	0	1
<i>Variables of Interest</i>				
Total successful appointments	3.561	1.847	0	8
Early intervention: Attended before October 14	.884	.321	0	1

Fuzzy regression discontinuity designs feature a cutoff that does not completely determine treatment status. These designs are essentially instrumental variables estimations where the cutoff does not completely determine treatment status but is sufficient to recover the exogeneity in the identification of an unbiased effect (Angrist & Pischke, 2008). Although traditional regression techniques can control variables that may confound treatment assignment, researchers remain uncertain if all selection biases are accounted for. The use of the fuzzy discontinuity in this study yields a more precise estimate of the treatment effect by using a variable outside of the control of the researchers and subjects to determine treatment status while also controlling for potential confounding variables. In our case, we use the running variable of timing of the first appointment to define early intervention and then exploit the fuzzy discontinuity to estimate a local average treatment effect of each additional coaching session on student outcomes for those who attend their first appointment before the cutoff.

For the purposes of this study, early interventions are defined as any student who successfully attended their first student success coaching session prior to the sixth week of their first fall 14-week semester, which began on October 14. We believe this date represents a meaningful discontinuity in the semester because starting in that week, students began receiving academic progress reports from their instructors, which alerts students if they are at-risk to fail any courses based on the coursework and assessments completed up to that point. One feature of the local average treatment effect is to estimate the treatment effect “local” to before the discontinuity, we can recover an effect for students who waited sufficiently long into the semester to make their

first appointment but were early enough to form relationships with their coach prior to the academic progress reporting period. If a student had one or more coaching sessions as of that date, they are assigned an indicator for early intervention in the dataset. Figure 1 is a distributional dotplot of the number of successful appointments by early intervention.

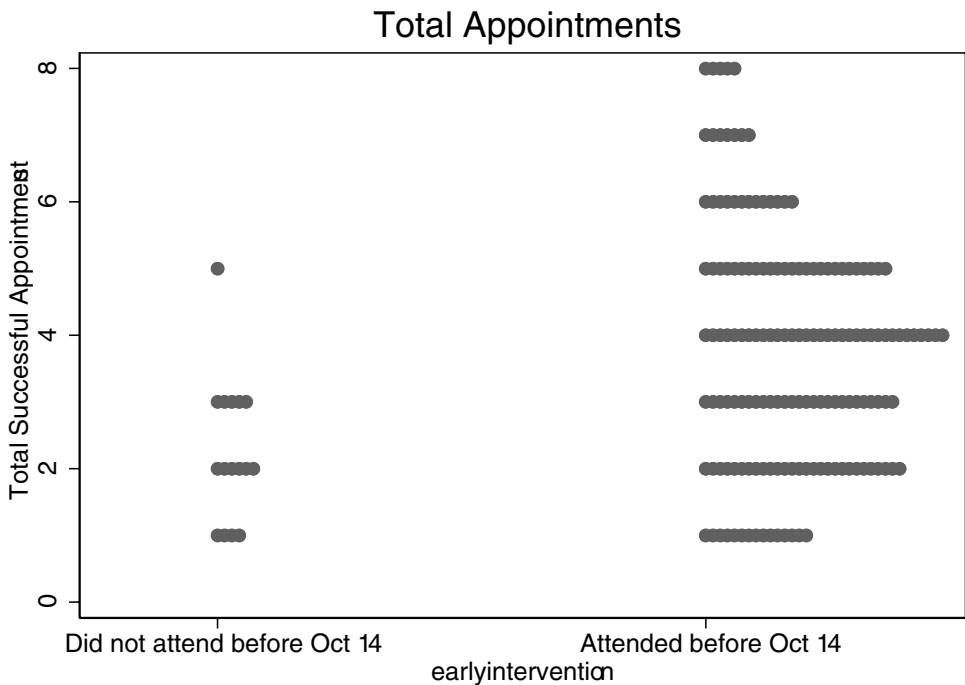
Formally, we estimate the following two-stage least squares (2SLS) equations:

$$z_i = \alpha_0 + x_i\alpha + \varepsilon_i$$

$$y_i = \beta_0 + \hat{z}_i\beta + \varepsilon_i$$

Where y is the student i 's grade point average or credits earned at the end of their first semester (term GPA or term credits), z is the number of student success coaching sessions that student i attended in their first semester, x is a binary indicator for student i 's participation in the early intervention student success coaching session, $\hat{z}$ is the predicted number of total successful appointments plugged in from coefficient α in the first stage regression, and β is the estimated local average treatment effect of $\hat{z}$ on y . By substituting in the information about differences in numbers of appointments across the discontinuity, we achieve a more accurate estimate of the effect of additional appointments for the group that attended early. It is called the local average treatment effect because the result is “local” to the students whose first attendance status is early in the semester, rather than for the whole population (Angrist & Pischke, 2008).

Figure 1. Distributional dot plot where each dot represents one student's total successful appointments during the whole semester. Students are grouped by the timing of their first appointment before and after the early intervention discontinuity, which was 6 weeks into their first semester.



In addition to these empirical models, a conditional model is estimated controlling for the high school GPA (centered and standardized with an indicator for missing values), SAT score (centered and standardized with an indicator for missing values), the student's gender, ethnicity (grouped into five categories), and an indicator of whether a student is the first in their family to attend college. This alternative model specification serves as a robustness check in the presence of differences, meeting a more relaxed “conditional mean independence” assumption (Wooldridge, 2010). If effects from the two models are similar, it improves our confidence in the findings. All models use heteroscedasticity robust standard errors to improve inference (Wooldridge, 2010).

This research design offers better estimates of the treatment effect than ordinary least squares (OLS) for at least two reasons. First, the research design resolves some selection bias because we observe a difference in the dosage of treatment across the discontinuity. Using OLS to estimate the effect of additional appointments may make additional coaching appear less effective because students who wait for their first appointment run out of time in the semester to fully develop relationships with coaches and realize the benefits of additional coaching, biasing the estimator toward zero. We also present the OLS estimates without instrumenting the total successful appointments variable, but instead include the early intervention indicator as an additional regressor for interested parties to demonstrate this difference. Second, by using the two-stage approach, we efficiently use the available data to both guide practice from the perspective of timing and frequency because the relationship between timing of first appointment appears to strongly predict the number of total appointments and allows us to make stronger claims about the effect for the population that attends earlier in the semester. Furthermore, we believe this difference is substantively related to our applications of Bandura's Social Cognitive Theory as an indicator of increasing self-efficacy for college success, a kind of proxy measure for the theory. This approach is similar to the approach in other research (Vanacore & Dahan, 2021).

To further rule out alternative explanations, we implemented a series of falsification tests of the data. First, we tested a predetermined variable (students' high school GPA) against our variables of interest, to ensure temporal consistency because participation should not be associated with previous grades (Cattaneo et al., 2020). Next, we limited our sample to those students who access their first meeting around a narrow band on either side of the discontinuity to ensure that the discontinuity is a valid source of exogeneity. Finally, we implemented tests of manipulation to demonstrate that students are not manipulating their treatment status on either side of the discontinuity (Cattaneo et al., 2018; McCrary, 2008).

Results

Assessing the Instrumental Variables

In the first stage of 2SLS models, students who participated in the early intervention attended an average of 3.815 student success coaching meetings conditional on covariates including a pell indicator, ethnicity, gender, first generation status, SAT score, and high school GPA ($t = 5.48, p < .001$), whereas the students who did not attend an early intervention (conditional on covariates) had an average of about 1.945 total student success coaching meetings ($t = 4.72, p < .001$). The instrumental variables literature recommends using an F -statistic greater than 10 to identify if an instrumental variable is sufficiently related to the endogenous variable (Angrist & Pischke, 2008; Staiger & Stock, 1997). In the conditional model, the F -statistic for the excluded instrument (early intervention) is 31.16 ($p < .001$), supporting early interventions as a strong instrument for total appointments, which is also supported by the distribution of appointments in Figure 1.

Local Average Treatment Effect

Table 2 presents the results of four models: the base OLS model, the base 2SLS model, the conditional OLS model, and the conditional 2SLS model. In the OLS models, the coefficient on total successful appointments is small. In the conditional model, it is marginally significant from

Table 2
Comparing Models Predicting First Semester GPA: Fuzzy Regression Discontinuities

	(1)	(2)	(3)	(4)
	OLS	OLS Conditional	2SLS	2SLS Conditional
Total Successful Appointments	0.040 (0.042)	0.058 (0.036)	0.257** (0.128)	0.320*** (0.114)
Attended before Oct 14	0.431* (0.257)	0.489** (0.222)		
Pell indicator = 1		−0.002 (0.123)		0.029 (0.139)
African American		−0.291* (0.153)		−0.356** (0.176)
Asian		0.004 (0.115)		0.047 (0.149)
Hispanic		−0.438** (0.184)		−0.562*** (0.206)
Other		−0.303** (0.143)		−0.568*** (0.204)
gender = = Female = 1		0.074 (0.147)		0.182 (0.177)
firstgen = = First Gen = 1		−0.019 (0.140)		−0.052 (0.159)
Mean-centered SAT Score (divided by 100)		0.102* (0.055)		0.155** (0.070)
Standardized values of (highschoolgpa)		0.245*** (0.069)		0.175** (0.078)
Indicator for missing SAT = 1		0.016 (0.254)		0.084 (0.202)
Indicator for missing HS GPA = 1		0.599*** (0.193)		0.689*** (0.190)
Constant	2.602*** (0.254)	2.568*** (0.238)	2.212*** (0.474)	2.059*** (0.444)
N	173	173	173	173
F	2.513	5.601		
chi2			4.009	54.362

zero ($t = 1.61$, $p = .109$). However, the discontinuity is represented by the indicator for early interventions. The result is significant and equal to almost half of a grade point difference ($\beta = .489$, $se = .222$, $p = .029$).

In the conditional 2SLS model, which uses the difference in the number of appointments across the discontinuity to estimate the effect of each additional meeting, the coefficient on total successful meetings is significant ($\beta = .320$, $se = .114$, $p = .005$). This effect is the average causal response for each additional meeting taken by the group of students who attended early interventions. The coefficient on total successful appointments in the OLS models assumes a constant treatment effect for each appointment, and the indicator for the discontinuity represents a shift in the intercept, but it obscures the difference in the number of appointments before and after the discontinuity. We can use our knowledge about the difference in the number of appointments between the group that attends early and later (1.8 meetings on average) and divide the OLS coefficient for early interventions to get a result that is roughly equal to the average causal response for each additional meeting in the 2SLS model. This difference demonstrates how the 2SLS estimates the dosage effect for additional meetings more accurately than an OLS model.

As a robustness check on the GPA model, Table 3 presents the same four models substituting the number of credits completed in the first semester as the response. These models also find that discontinuity is present, both in the OLS model with the fuzzy discontinuity included as an indicator and in the 2SLS with the fuzzy discontinuity estimated as an instrumental variable. Like the GPA models, the OLS model finds the shift in the intercept for early intervention participation to be roughly twice as large as the effect estimated in the 2SLS model.

Falsification Tests

As a falsification test, we regressed high school GPA (instead of first semester GPA) on the indicator for early intervention and the total successful appointments. This tests the assumption that the early intervention only effects the intended outcome and not pre-determined variables. There are no relationships for coefficients on the early intervention ($\beta = -.105$, $t = -.43$, $p = .667$) or on total appointments ($\beta = .052$, $t = 1.22$, $p = .223$). We included additional mean comparison tests in Appendix Table A1, establishing group equivalence across the discontinuity, which improves our confidence that the instrumental variable meets the exclusion restriction.

In addition, we ran a simple t -test for first semester GPA on a narrow band of data including 4 weeks on either side of the discontinuity of the running variable (timing of first appointment). The difference in GPA is significant ($d = .696$, $t(61) = 3.024$, $p = .004$). Although this result is larger than the results presented here, it is consistent with that finding and improves our confidence in the results presented.

Furthermore, we implemented a manipulation test using a procedure described by Cattaneo et al. (2018) that uses local polynomial-based density estimation to determine if there is evidence of systematic manipulation around the discontinuity. A key assumption of regression discontinuity is that subjects cannot manipulate their own treatment assignment at the cutoff, so a significant result on this test would mean a violation of that assumption (McCrary, 2008). We failed to reject the null hypothesis of no manipulation ($t = .116$, $p = .908$). Refer to Appendix Figure A1 for additional detail.

Table 3

Comparing Models Predicting First Semester Credits Completed: Fuzzy Regression Discontinuities

	(1)	(2)	(3)	(4)
	OLS	OLS Conditional	2SLS	2SLS Conditional
Total Successful Appointments	0.045 (0.100)	0.072 (0.099)	0.691* (0.356)	0.886** (0.370)
Attended before Oct 14	1.275* (0.698)	1.508** (0.667)		
Pell indicator = 1		−0.325 (0.346)		−0.205 (0.390)
African American		−0.448 (0.569)		−0.632 (0.576)
Asian		0.165 (0.518)		0.279 (0.590)
Hispanic		−0.307 (0.499)		−0.712 (0.630)
Other		1.049* (0.571)		0.333 (0.746)
gender = = Female = 1		−0.101 (0.368)		0.248 (0.455)
firstgen = = First Gen = 1		0.567 (0.379)		0.480 (0.422)
Mean-centered SAT Score (divided by 100)		0.708*** (0.185)		0.882*** (0.205)
Standardized values of (highschoolgpa)		0.164 (0.205)		−0.036 (0.231)
Indicator for missing SAT = 1		0.495 (0.577)		0.682 (0.796)
Indicator for missing HS GPA = 1		−0.863 (0.605)		−0.575 (0.697)
Constant	14.018*** (0.652)	13.599*** (0.738)	12.856*** (1.295)	11.980*** (1.440)
N	175	175	175	175
F	2.274	2.861		
chi2			3.761	29.483

Note: Standard errors in parentheses * $p < .1$, ** $p < .05$, *** $p < .01$.

Discussion

The results in this study point to the effectiveness of the coaching program when students participate at an early point in the semester. When serving as an early intervention in students' experiences, coaches build relationships as they discuss effective strategies for student success, help students navigate the campus and the institutional culture, and serve as an ongoing support system that students can rely on throughout the first semester. Additionally, we find that the ongoing relationship between a coach and student to be important for effects to be maximized. Early and frequent coaching appears to enable positive outcomes for students because the timing allows coaches to not only help students prepare and respond to obstacles, but also prevent challenges from occurring altogether. In this case, timely coaching sessions facilitate the use of proactive strategies (i.e., course progress monitoring, practice/demonstration of learning strategies) that promote academic success.

Our overarching theory guiding our coaching work is the social cognitive theory (Bandura, 1986). College students benefit from coaching because they are offered support in developing behavioral capabilities for success in college as presented in the procedures of this study. Students' coaches reinforce concepts that help them develop skills and knowledge for success in college. As observed with positive outcomes related to increases in the dosage of coaching, the repeated reinforcement of coaching toward positive behaviors develops students' self-efficacy for personal success in college and enhances their expectations about how to achieve goals set for themselves, which results in positive outcomes such as increases in GPA and credits completed (Bandura, 1997). To that end, we argue that our dosage effects are meaningful proxies of the socially interdependent and richly contextualized agentic perspectives that are embedded in Social Cognitive Theory (Bandura, 2001).

We believe that having an early intervention is an effective strategy for delivering information about success strategies. We argue that timing and frequency of coaching measured in this study may also serve as a good proxy measure for these practices similar to the approach in Vanacore and Dahan (2021), especially given the procedures for the coaching model described extensively in the methodology. Previous scholarship has pointed to the importance of developing success strategies with students as they make the transition from high school to college (Nagaoka et al., 2013). As found in research on academic coaching, programs can promote academic skills that foster student success (Capstick et al., 2019; Ralston & Hoffshire, 2017; Vanacore & Dahan, 2021). These skills include time management, study habits, self-efficacy, and academic hope (Dill et al., 2014; Hansen et al., 2014; Tuckman & Kennedy, 2011). These skillsets form a system of academic self-regulation (Zimmerman & Schunk, 2001), which draws heavily on Bandura's theories of self-efficacy (Bandura, 1997). Additionally, success coaches help students navigate the first year, and as a result, these students are more academically integrated into the college environment (Tinto, 1993). Students who establish relationships with coaches in their first year are able to draw on the support of coaches throughout their collegiate experience (Alzen et al., 2021).

In this study, we provide a detailed description of our coaching program, following the example of Alzen et al. (2021). The success coaching program described here is consistent with models described in the emerging literature regarding the practice (Robinson, 2015; Sepulveda, 2020), and this research contributes more detail to this growing field. Furthermore, although our study does not directly measure self-efficacy or other constructs, by providing readers details of the coaching procedures, practitioners have a roadmap for making the theory-practice connection

between the practical importance of timing and frequency and more complex theoretical mechanisms that these behaviors are approximating in our study (Vanacore & Dahan, 2021).

The effect sizes of our results are consistent with the findings from previous research, which further increases our confidence in the validity of the results. Our finding of 0.2–0.3 grade points on a 4-point scale is roughly the same as the 4–8 point difference on a 100-point scale uncovered via randomized control trial by Oreopoulos and Petronijevic (2018). As noted earlier, the research by Oreopoulos and Petronijevic (2018) featured peer coaches, future research may further examine the difference between results achieved by professional coaches vs. peer coaches, and multisite research may offer the best opportunity for this kind of research.

In addition, the findings contribute to our understanding of when coaching is most effective. Our results support Bettinger and Baker's (2014) finding that coaching was more effective when a student responded to their coach's outreach by the first week of classes. Additionally, our findings are consistent with previous literature regarding dosage effects where additional coaching is associated with better outcomes (Alzen et al., 2021; Bettinger & Baker, 2014; Linkow et al., 2017; Pechac & Slantcheva-Durst, 2021). Furthermore, by defining and contributing the dose-response terminology to the field of student affairs, we hope that practitioners consider dosage effects when designing programs and researchers continue to specifically explore these relationships in coaching and other student-focused activities.

Implications for Practice

Our findings support two conclusions regarding the practical implications of early and frequent success coaching. First, students benefit most from participation in success coaching if the first appointment occurs at an early enough point in the semester. Early engagement with a coach is antecedent for both relationship-building and the discussion of success strategies to be effective and is therefore theoretically and practically relevant to establishing relationships that promote self-efficacy. With the understanding that establishing personal relationships and a sense of belonging on campus enhances students' likelihood of success, it becomes that much more important that coaching is seen as a conduit for connection within the college environment. We recommend that coaches promote themselves—early and often—as advocates and advertise the potential benefits of services so that students are drawn to explore the possibilities of coaching.

Leading up to and during the first meeting, establishing a positive rapport and providing students with tangible skills or knowledge can help them to immediately understand the value of coaching. It is presumed that upon witnessing positive impacts from the coaching intervention, students will be more likely to further invest in the relationship with their coach and thus attend sessions more frequently. We believe it is the strength of the initial bond that influences students' receptivity to coaching strategies and thus, improved success skills. We recommend that programs aim to serve as many students in the first 6 weeks of classes as possible. Patterns in our data found that most of the students served had scheduled their first interaction with a coach in the period bounded by 2 weeks prior to and after the start of classes, but as we demonstrate in this study, even those served as late as 4 or 5 weeks into the semester benefited from the program.

Second, we observed that students who participated in the early intervention access an average of two additional meetings compared to the group that waits until the midsemester to access their first meeting. We argued that additional coaching is evidence of increasing self-efficacy for college success and is as theoretically relevant as it is practically relevant. Our findings

suggest a guideline of between four and six meetings across a 15-week semester are necessary for success coaching to be most effective. Although four to six meetings are optimal, our evidence supports an additive dosage effect that increases as the number of appointments increases for students that access coaching early in the semester.

Limitations

The results presented here support the conclusion that success coaching is most effective when delivered early in the semester and frequently throughout the semester, but this study has limitations. First, our data did not include variables for college majors, which is a potential source of confounding, and we can make no statements regarding differences by that factor. There may be other unobservable factors that further confound our results.

A second limitation is approximately 50% of students who participated in success coaching declined consent to participation in this study. Although these sample limitations raise issues related to confounding and the potential for attrition bias (What Works Clearinghouse, 2020), we are confident in our results for at least three reasons:

1. the differential rates of attrition for the groups within the narrow band around the discontinuity are equivalent (see Appendix Figure A1);
2. the baseline equivalence of groups is not compromised in either the full sample or the sample around the narrow band (see Appendix Table A1); and,
3. the evidence presented in the falsification tests earlier further support our overall findings.

Another major limitation in this data is the lack of a description of the success coaching program as experienced by the students that might yield additional understanding of why early and frequent coaching works. We acknowledge reconciling this limitation would add additional support for our use of the dosage measures as proxies for complex, socially interdependent, theoretical constructs. At this stage in our research, we are testing the theory that additional meetings are most effective when students start their relationship with their coach early and continue throughout the semester, but this research is our first step in a multistage mixed methods inquiry. Future directions for this research include interviewing students with representative values on various characteristics in our regression models to begin constructing rich descriptions of the coaching process (Barton, 2023; Creswell & Plano Clark, 2017). Additionally, our study is limited in understanding how students benefit from participation in coaching. We recommend more research on student identities and how participation in coaching differs among diverse students. Such contributions would help practitioners understand who benefits most from coaching, if coaching benefits are equitable across identities, and how students from diverse backgrounds benefit from coaching.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Supplementary Material

Supplemental data for this article can be accessed online at <https://doi.org/10.1080/19496591.2023.2247374>.

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